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Gravity plays an important role in the obtained solidification microstructures mainly by affecting the melt flow mechanisms as well as the solute diffusion processes. Since these effects remain present even under reduced gravity environments, we address inverse design solidification problems that result in desired microstructures at various gravity levels.
Solidification control is achieved with proper design of boundary thermal conditions as well as with the application of electromagnetic fields. Designing appropriate strengths and direction of magnetic fields simultaneously with continuous boundary heating/cooling conditions for various strengths of gravity will further enhance the success of the solidification design process.

These design problems are posed as inverse problems with over-specified thermal conditions (temperature and flux) on

Task Description:	one part of the boundary and no-thermal conditions on another part of the boundary. They are stated as functional optimization problems in the L_2 space. The exact gradient of the cost functional is obtained via the solution of an adjoint continuum problem. A sensitivity problem is also defined and is used in the implementation of the conjugate gradient method. Conjugate gradient techniques and preconditioning are considered in the implementation of these optimization algorithms. Deformable-Spatial-Domain/Stabilized-Space-Time finite element techniques are the methods of choice for the solution of the direct, sensitivity and adjoint problems. Regularization techniques are also considered due to the ill-posed nature of the proposed models. All proposed computations are implemented on the Intel-based cluster supercomputer at the Cornell Theory Center. The developed techniques are used to actively control natural convection and other secondary effects of gravity on the obtained solidification microstructures. New accurate mathematical designs for microstructure control are being developed and compared with design solutions currently available through experimentation or trial and error techniques. Two and three-dimensional processes are considered in our simulations for metal alloys and semiconductor materials for which the thermo-physical properties are well documented. Development of the proposed inverse techniques and strong interactions with process design practitioners in industry and NASA laboratories would enhance our ability to develop new advanced directional solidification process designs for cast microstructure control. The main objective of the proposed work is to design and test computational techniques that can be used to control the developed microstructures in directional solidification processes. Our main design variables are the thermal mold/furnace conditions as well as the strength and orientation of the electromagnetic fields. We select these continuous variables such that a desired growth velocity V and temperature field (e.g. a gradient G) is achieved near the freezing interface that correspond to desired microstructures. In our preliminary work, we are interested to design processes with spatially uniform growth velocity under stable growth conditions. The techniques under investigation are crucial for the development of the next generation of furnaces for controlled crystal growth. In particular, the availability of mathematical models for solidification control via optimally designed thermal boundary fluxes and electromagnetic fields is essential for the development of multiple zone-instrumented furnaces. In this work, we are using rigorous inverse problem theory for the design and control of macroscopic quantities (i.e. of G and V) that are directly linked with the stability of the melt/crystal interface and the morphology on the obtained microstructures. Implementation of such computational models: (a) will provide the mathematical tools necessary for the development and use of automated furnaces that for various materials and processing conditions can control the growth velocity and temperature field near the interface, (b) will provide the means to design directional casting processes and Bridgman crystal growth processes that lead to desired cast structures and (c) will motivate new experimental designs where active control of the melt flow may lead to a better understanding of the effects of convection and thermal/mass diffusion to the obtained microstructures. To the best of our knowledge, this is the first time that functional adjoint techniques are applied for the solution of inverse coupled heat/mass/flow transfer problems. As such, the implications of the proposed work are beyond the area of solidification processing. In particular, such inverse techniques can be applied to a variety of design, control and parameter identification problems for processes involving transport mechanisms. Program success will spur other researchers to investigate similar approaches in the design of other materials processes and areas of science & engineering. The design techniques under development provide the first step towards the real time control of solidification processes under uncertainty. The complex design solutions under development and the associated sensitivity analyses can identify effective and ineffective control and actuation mechanisms and design objectives before they are implemented. This information can be used to design low-order controllers and to determine effective locations to place sensors and actuators for real time control of the process. In addition, the present complex design solutions are essential in producing snapshots for reduced-order solutions, which in turn can be used to design controllers that can be applied in real time. Of course, a fundamental use of complex design solutions is for passive (off-line) optimal design. Many process features cannot be varied, but are fixed, e.g., container geometries. Often system performance can be improved dramatically by optimally designing these fixed features.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Many of the developed techniques can be used as alternate means to microgravity for control of crystal growth processes. For example, we have shown that the use of properly selected non-uniform magnetic fiels can lead to solidification growth conditions that are similar to those obtained under microgravity conditions. A combined reduced gravity environment and proper control of the process via furnace or magnetic field design will lead to better crystal growth processes resulting in crystals of reduced defects and increased quality. Finally, the techniques developed for the control of the growth non conducting materials may be important in the crystallization of proteins or other biomolecules.
	(1) Using magnetic field gradients to control the directional solidification of alloys and the growth of single crystals There are various magnetic field based approaches to melt flow control that are used commercially to improve the quality of a crystal grown from the melt. Most of them are based on either a uniform or a rotating magnetic field. In work completed this year (Journal of Crystal Growth, B. Ganapathysubramanian and N. Zabaras, Vol. 270/1-2, pp. 255-272, 2004), the effect of applying a suitably positioned magnetic field gradient on the growth of single crystals, particularly the interface curvature, growth rate and dopant distribution was investigated. Comparison between crystal growth under the influence of a uniform magnetic field and a magnetic field gradient brings out the significant advantages of the developed method. The use of a magnetic gradient was shown to give a uniform dopant distribution and produce a planar interface. Both directional solidification and horizontal Bridgeman growth processes were considered. The ease of control as well as the commercial feasibility of such a technique was emphasized. Finally, its promising role in biological macromolecular growth was addressed. (2) Control of solidification of non-conducting materials using tailored magnetic fields The structure of fluid flow in a solidifying melt plays a critical role in the quality/properties of the solid. It follows that by controlling the flow behavior, the final solidified material can be suitably affected. Based on this concept, there exist different magnetic field approaches to melt flow control that are commercially used for improving the quality of the

Task Progress:	solidified product. Most of them depend on the variation of the Lorentz force for suppression of flow and are limited to conducting materials alone. However, simultaneous application of a magnetic gradient gives rise to the Kelvin force that can be used to affect the melt flow of any material, conducting or otherwise, resulting in better control of the crystal quality. In this work (B. Ganapathysubramanian and N. Zabaras, Journal of Crystal Growth, in press), a computational method for the design of solidification of a non-conducting material is addressed such that diffusion-dominated growth is achieved by the suppression of convection. The control parameter in the design problem is the time history of the imposed magnetic field. A steady, constant magnetic gradient is also maintained during the process. The design problem is posed as an unconstrained optimization problem with the cost functional taken as the square of the L2 norm of an expression representing the deviation of the velocity field in the melt region from conditions corresponding to convection-less growth. The adjoint method for the inverse design of continuum processes is adopted. The cost functional minimizing process is realized through the use of the conjugate gradient method via the finite element solutions of the continuum direct, sensitivity and adjoint problems. Examples of designing the time history of the imposed magnetic field for the directional growth of various non-conducting materials are presented to demonstrate the developed formulation. To the best of our knowledge, this is the first rigorous computational design study of the controlled solidification of non-conducting materials using magnetic field gradients. (3) Control of solidification of conducting materials using tailored magnetic fields Solidification from the melt to near net shape is a commonly used manufacturing technique for conducting materials. The fluid flow patterns in the melt affect the quality and properties of the final product. By controlling the flow behavior, the final solidified material can be suitably affected. In this context, magnetic fields offer a promising means of controlling the solidification process. Most of the magnetic field approaches to melt flow control rely on the application of a constant magnetic field. A constant magnetic field results in the Lorentz force which is used to damp and control the flow. However, simultaneous application of a magnetic gradient results in the Kelvin force along with the Lorentz force. This can be used for better control of the melt flow resulting in higher crystal quality. In this completed work (B. Ganapathysubramanian and N. Zabaras, Int. J. of Heat and Mass Transfer, in press), a computational method for the design of solidification of a conducting material is addressed. The control parameter in the design problem is the time history of the imposed magnetic field. A steady, constant magnetic gradient is also maintained during the process. The design problem is posed as an unconstrained optimization problem with the cost functional taken as the square of the L2 norm of an expression representing the deviation of the velocity field in the melt region from conditions corresponding to convection-less growth. The adjoint method for the inverse design of continuum processes is adopted and the corresponding continuum adjoint equations derived. The cost functional minimizing process is realized through the use of the conjugate gradient method via the finite element solutions of the continuum direct, sensitivity and adjoint problems. Examples of designing the time history of the imposed magnetic field for the directional growth of various semi-conductors were presented to demonstrate the developed formulation. We believe that with the completion of this work we accomplish the main proposed objectives of this NASA-project – mainly the development of computational tools for the design of directional solidification processes via controlled furnace conditions and electromagnetic fields.

Bibliography Type:	Description: (Last Updated: 01/31/2005)
Articles in Peer-reviewed Journals	Ganapathysubramanian, B.; Zabaras, N. "Using magnetic field gradients to control the directional solidification of alloys and the growth of single crystals" Journal of Crystal Growth, Vol. 270/1-2, pp. 255-272 , Dec-2004
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Papers from Meeting Proceedings	Zabaras, N.; Ganapathysubramanian, B. "Melt flow control using magnetic fields and magnetic field gradients" invited presentation at the symposium on `Computational electro-magneto-fluid-dynamics' (G. Dulikravich, organizer), Sixth World Congress on Computational Mechanics (WCCM VI) in conjunction with the Second Asian-Pacific Congress on Computational Mechanics (APCOM'04), Beijing, China, September 5-10, 2004 Sep-2004
Presentation	Zabaras, N. "On the control of crystal growth processes using magnetic fields and magnetic field gradients" Invited keynote lecture at FLOWCOMAG, International workshop on Flow Control by Tailored Magnetic Fields, Dresden, Germany, April 1-2 Apr-2004